

Digital Design and Computer Organisation Laboratory

UE23CS251A

3rd Semester, Academic Year 2024-25

Date:12/09/2024

Name: Dipak Chaudhari	SRN: PES1UG24CS804	Section K
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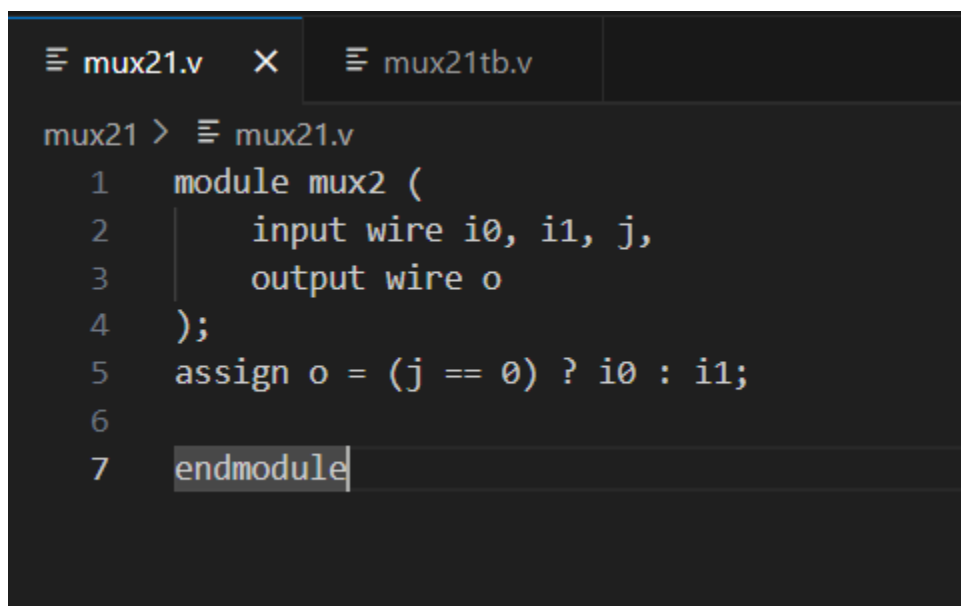
Week# 4

Program Number: 1

Title:

WRITE A VERILOG PROGRAM TO MODEL A 2:1 MUX

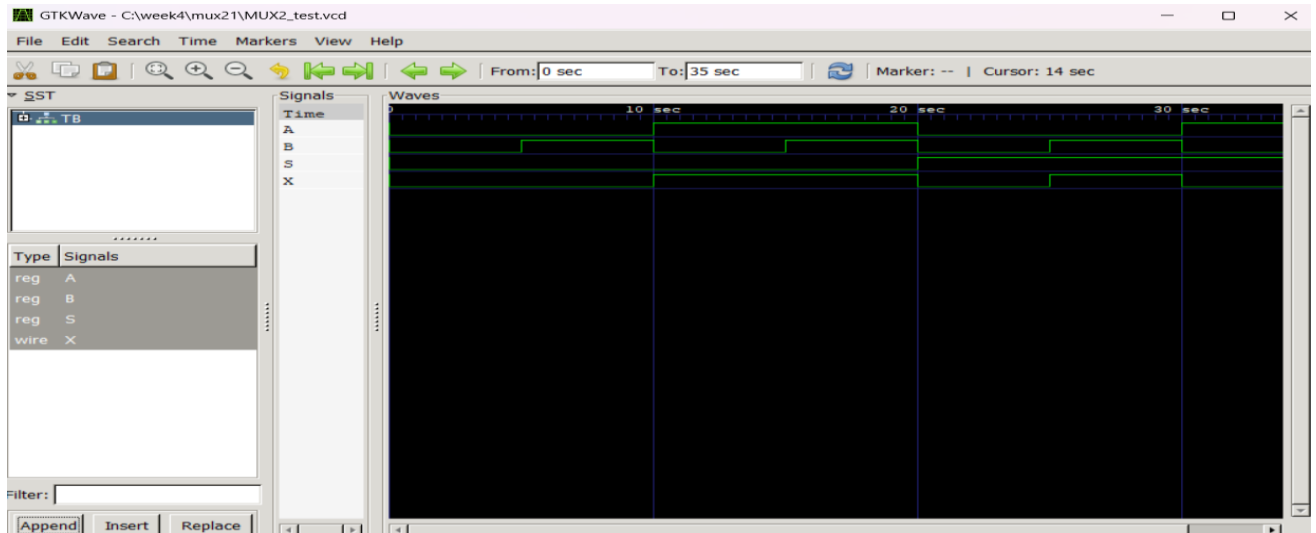
- I. Verilog Code Screenshot
- II. Verilog VVP Output Screen Shot
- III. GTKWAVE Screenshot
- IV. Circuit Diagram

A screenshot of a Verilog code editor with a dark background. At the top, there are two tabs: 'mux21.v' and 'mux21tb.v'. The 'mux21.v' tab is active, showing the following code:

```
1 module mux2 (  
2     input wire i0, i1, j,  
3     output wire o  
4 );  
5 assign o = (j == 0) ? i0 : i1;  
6  
7 endmodule
```

```
mux21.v  mux21tb.v X
mux21 > mux21tb.v
1  module TB;
2      reg A, B, S;
3      wire X;
4      mux2 newMUX(.i0(A), .i1(B), .j(S), .o(X));
5
6      initial begin
7          S = 1'b0; A = 1'b0; B = 1'b0;
8          #5 S = 1'b0; A = 1'b0; B = 1'b1;
9          #5 S = 1'b0; A = 1'b1; B = 1'b0;
10         #5 S = 1'b0; A = 1'b1; B = 1'b1;
11         #5 S = 1'b1; A = 1'b0; B = 1'b0;
12         #5 S = 1'b1; A = 1'b0; B = 1'b1;
13         #5 S = 1'b1; A = 1'b1; B = 1'b0;
14         #5 S = 1'b1; A = 1'b1; B = 1'b1;
15     end
16     initial begin
17         $monitor("Time = %0t: A = %b, B = %b, S = %b, X = %b", $time, A, B, S, X);
18     end
19     initial begin
20         $dumpfile("MUX2_test.vcd");
21         $dumpvars(0,TB);
22     end
23 endmodule
```

```
C:\week4\mux21>vvp test
VCD info: dumpfile MUX2_test.vcd opened for output.
Time = 0: A = 0, B = 0, S = 0, X = 0
Time = 5: A = 0, B = 1, S = 0, X = 0
Time = 10: A = 1, B = 0, S = 0, X = 1
Time = 15: A = 1, B = 1, S = 0, X = 1
Time = 20: A = 0, B = 0, S = 1, X = 0
Time = 25: A = 0, B = 1, S = 1, X = 1
Time = 30: A = 1, B = 0, S = 1, X = 0
Time = 35: A = 1, B = 1, S = 1, X = 1
```



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Week# 4

Program Number: 2

Title:

WRITE A VERILOG PROGRAM TO MODEL A 4:1 MUX

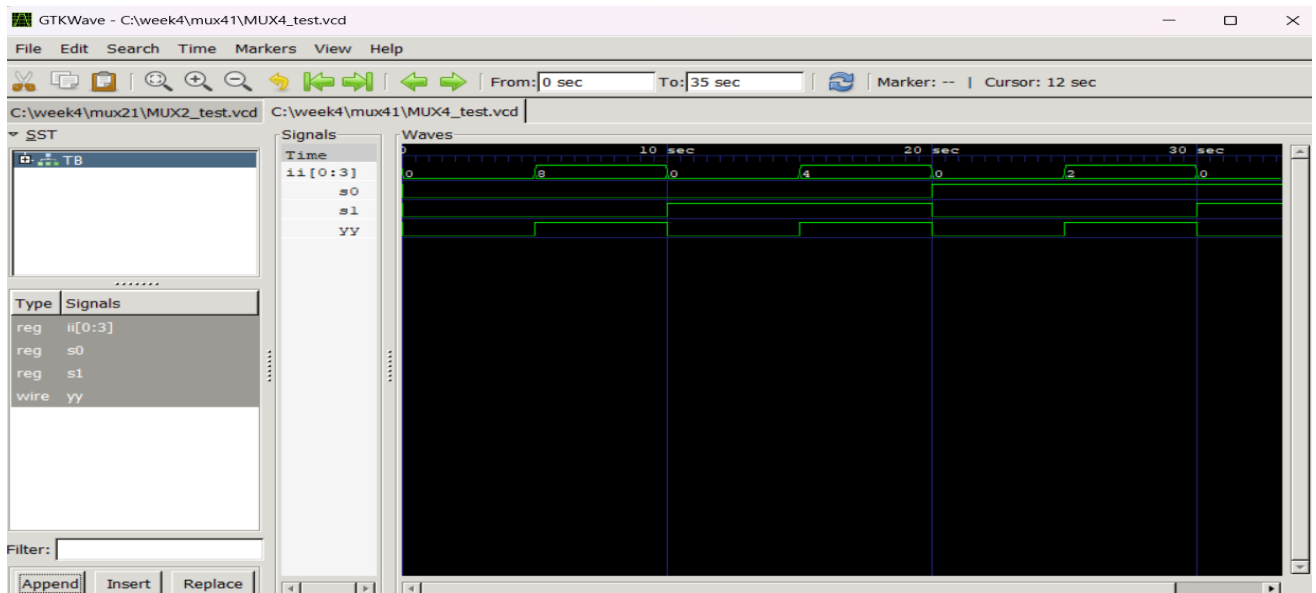
- I. Verilog Code Screenshot
- II. Verilog VVP Output Screen Shot
- III. GTKWAVE Screenshot

IV. Circuit Diagram

```
mux41.v  X  mux41tb.v
mux41 >  mux41.v
1  |
2  | module mux4(input wire [0:3] i, input wire j0, j1, output wire o);
3  |     wire [0:1] s;
4  |     assign s = j1 ? {j0, 1'b1} : {j0, 1'b0};
5  |
6  |     assign o = (s == 2'b00) ? i[0] :
7  |                 (s == 2'b01) ? i[1] :
8  |                 (s == 2'b10) ? i[2] :
9  |                 i[3];
10 | endmodule
```

```
mux41.v  X  mux41tb.v  X
mux41 >  mux41tb.v
1  | module TB;
2  |     reg [0:3] ii;
3  |     reg s0; reg s1;
4  |     wire yy;
5  |     mux4 newMUX(.i(ii), .j0(s0), .j1(s1), .o(yy));
6  |     initial begin
7  |         ii = 4'b0000; s0=1'b0; s1=1'b0;
8  |         #5      ii = 4'b1000; s0=1'b0; s1=1'b0;
9  |         #5      ii = 4'b0000; s0=1'b0; s1=1'b1;
10 |         #5      ii = 4'b0100; s0=1'b0; s1=1'b1;
11 |         #5      ii = 4'b0000; s0=1'b1; s1=1'b0;
12 |         #5      ii = 4'b0010; s0=1'b1; s1=1'b0;
13 |         #5      ii = 4'b0000; s0=1'b1; s1=1'b1;
14 |         #5      ii = 4'b0001; s0=1'b1; s1=1'b1;
15 |     end
16 |     initial begin
17 |         $monitor("Time = %0t: ii = %b, s0 = %b, s1 = %b, yy = %b", $time, ii, s0, s1, yy);
18 |     end
19 |     initial begin
20 |         $dumpfile("MUX4_test.vcd");
21 |         $dumpvars(0, TB);
22 |     end
23 | endmodule
```

```
C:\week4\mux41>vvp test
VCD info: dumpfile MUX4_test.vcd opened for output.
Time = 0: ii = 0000, s0 = 0, s1 = 0, yy = 0
Time = 5: ii = 1000, s0 = 0, s1 = 0, yy = 1
Time = 10: ii = 0000, s0 = 0, s1 = 1, yy = 0
Time = 15: ii = 0100, s0 = 0, s1 = 1, yy = 1
Time = 20: ii = 0000, s0 = 1, s1 = 0, yy = 0
Time = 25: ii = 0010, s0 = 1, s1 = 0, yy = 1
Time = 30: ii = 0000, s0 = 1, s1 = 1, yy = 0
Time = 35: ii = 0001, s0 = 1, s1 = 1, yy = 1
```



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Week# 4

Program Number: 3

Title:

WRITE A VERILOG PROGRAM TO MODEL A 2:1 DEMUX

- I. Verilog Code Screenshot
- II. Verilog VVP Output Screen Shot
- III. GTKWAVE Screenshot
- IV. Circuit Diagram

```
demux2.v X tb.v
demux > demux2.v
1  module demux2 (
2      input wire in,
3      input wire sel,
4      output wire y0,
5      output wire y1
6  );
7
8      assign y0 = (~sel) ? in : 1'b0;
9      assign y1 = sel ? in : 1'b0;
10
11  endmodule
12
```

```
demux2.v  tb.v  X
demux > tb.v
1  module TB;
2
3      reg in, sel;
4      wire y0, y1;
5
6      demux2 demux2 (
7          .in(in),
8          .sel(sel),
9          .y0(y0),
10         .y1(y1)
11     );
12
13     initial begin
14         $dumpfile("demux1to2_test.vcd");
15         $dumpvars(0, TB);
16     end
17
18     initial begin
19         $monitor($time, " in = %b, sel = %b, y0 = %b, y1 = %b", in, sel, y0, y1);
20     end
21
22     initial begin
23         in = 1'b0; sel = 1'b0; #5;
24         in = 1'b1; sel = 1'b0; #5;
25         in = 1'b0; sel = 1'b1; #5;
26         in = 1'b1; sel = 1'b1; #5;
27         $finish;
28     end
29
30 endmodule
31
```

```
C:\week4\demux>vvp test
VCD info: dumpfile demux1to2_test.vcd opened for output.
      0 in = 0, sel = 0, y0 = 0, y1 = 0
      5 in = 1, sel = 0, y0 = 1, y1 = 0
     10 in = 0, sel = 1, y0 = 0, y1 = 0
     15 in = 1, sel = 1, y0 = 0, y1 = 1
tb.v:27: $finish called at 20 (1s)
```

